

JOSE M. ALVAREZ

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cc-jalvarez.github.io

CURRENT POSITION

Santander AI Lab Senior AI Researcher	Mar 2026 - Madrid, ES
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EDUCATION

Scuola Normale Superiore Ph.D. in Data Science Advisor: Salvatore Ruggieri Dissertation: “Causality for Fair Machine Learning” Committee: Mykola Pechenizkiy , Joshua Loftus , Eirini Ntoutsis	2020 - 2024
Toulouse School of Economics M.Sc. in Econometrics and Empirical Economics	2015 - 2017
University of Florida B.A. in Economics, <i>summa cum laude</i> B.A. in History	2011 - 2015
Lester B. Pearson United World College of the Pacific International Baccalaureate	2009 - 2011

AWARDS & HONORS

Best Reviewer , International Conference on Machine Learning	2024
Marie Skłodowska-Curie Fellowship (NoBIAS ITN), European Commission	2020 - 2023
Phi Beta Kappa , University of Florida	2015
Honors Program , University of Florida	2011 - 2015
Davis United Word College Scholar (full-ride), University of Florida	2011 - 2015
United Word College Scholarship (full-ride), Lester B. Pearson UWC	2009 - 2011

PUBLICATIONS & MANUSCRIPTS

Research interests: responsible AI, causal algorithmic fairness, human-in-the-loop

See [Google Scholar](#) for an up to date list; * shared first author

Journals

1. **J. M. Álvarez** and S. Ruggieri. Mutatis mutandis: Revisiting the comparator in discrimination testing. *Computational Intelligence*, 42(4):e70238, 2026. [[paper](#)] [[code](#)]
2. **J. M. Álvarez** and S. Ruggieri. Counterfactual situation testing: From single to multidimensional discrimination. *J. Artif. Intell. Res.*, 82:2279–2323, 2025. [[paper](#)] [[code](#)]
3. **J. M. Álvarez**, A. Bringas-Colmenarejo, A. Elobaid, S. Fabbriizzi, M. Fahimi, A. Ferrara, S. Ghodsi, C. Mougan, I. Papageorgiou, P. Reyero, et al. Policy advice and best practices on bias and fairness in AI. *Ethics and Information Technology*, 26(2):31, 2024. [[paper](#)]

4. M. Lazzari, **J. M. Álvarez**, and S. Ruggieri. Predicting and explaining employee turnover intention. *Int. J. Data Sci. Anal.*, 14(3):279–292, 2022. [\[paper\]](#)

Conferences

1. L. Bothmann, P. A. Boustani, **J. M. Álvarez**, G. Casalicchio, B. Bischl, and S. Dandl. Privilege scores. In *FAccT*, pages 4293–4341. ACM, 2026. [\[paper\]](#) [\[code\]](#)
2. **J. M. Álvarez** and S. Ruggieri. Toward a causal framework for modeling perception. *Proceedings of the AAAI/ACM Conference on AI, Ethics, and Society*, 8(1):166–178, 2025. [\[paper\]](#)
3. F. Palomba*, A. Pugnana*, **J. M. Álvarez**, and S. Ruggieri. A causal framework for evaluating deferring systems. In *AISTATS*, pages 2143–2151. PMLR, 2025. [\[paper\]](#) [\[code\]](#)
4. **J. M. Álvarez**, A. Mastropietro, and S. Ruggieri. The initial screening order problem. In *WSDM*, pages 165–174. ACM, 2025. [\[paper\]](#) [\[code\]](#)
5. **J. M. Álvarez** and S. Ruggieri. Counterfactual situation testing: Uncovering discrimination under fairness given the difference. In *EAAMO*, pages 2:1–2:11. ACM, 2023. [\[paper\]](#) [\[code\]](#)
6. **J. M. Álvarez***, K. M. Scott*, B. Berendt, and S. Ruggieri. Domain adaptive decision trees: Implications for accuracy and fairness. In *FAccT*, pages 423–433. ACM, 2023. [\[paper\]](#) [\[code\]](#)
7. C. Mougán, **J. M. Álvarez**, S. Ruggieri, and S. Staab. Fairness implications of encoding protected categorical attributes. In *AIES*, pages 454–465. ACM, 2023. [\[paper\]](#) [\[code\]](#)
8. S. Ruggieri, **J. M. Álvarez**, A. Pugnana, L. State, and F. Turini. Can we trust fair-AI? In *AAAI*, pages 15421–15430. AAAI Press, 2023. [\[paper\]](#)

Workshops

1. **J. M. Álvarez**. Implementing Causal Perception: Competing SCMs and Situated Fairness. Forthcoming in *BIAS @ ECML/PKDD*, 2026. [\[paper\]](#)
2. **J. M. Álvarez**, A. Bringas-Colmenarejo, A. Elobaid, S. Fabbriizzi, M. Fahimi, A. Ferrara, S. Ghodsi, C. Mougán, I. Papageorgiou, P. R. Lobo, M. Russo, K. M. Scott, L. State, X. Zhao, and S. Ruggieri. Policy advice and best practices on bias and fairness in artificial intelligence: An extended abstract. In *EWAF*, volume 3908 of *CEUR Workshop Proceedings*. CEUR-WS.org, 2024.

Other

1. **J. M. Álvarez**, A. Fabris, C. Heitz, C. Hertweck, M. Loi, and M. Zehlike, editors. *Proceedings of the 2nd European Workshop on Algorithmic Fairness, Winterthur, Switzerland, June 7th to 9th, 2023*, volume 3442 of *CEUR Workshop Proceedings*, 2023. CEUR-WS.org

Working Papers

1. Investigating Perception in LLM-Based Causal Graph Elicitation with J. Arias-Rodriguez and J. M. de la Chica-Rodriguez. *Under submission*.
2. Count Time Series Change Point Detection with W. Verbeke, J. Davis. *In preparation*.
3. The Weighted Partial Dependence Plot with A. Pugnana, S. Ruggieri. *In preparation*.

RESEARCH VISITS

KBC, Advanced Data Analytics and Modeling
Host: Dr. Barak Chizi

Nov 2024 - Feb 2026
 Leuven, BE

SCHUFA Holding AG, Innovation & Research Team
Host: Dr. Klaus Broelemann

Jul - Aug 2023
 Wiesbaden, DE

Saarland University , Probabilistic Machine Learning Group <i>Host: Prof. Isabel Valera</i>	Aug 2023 Saarbrücken, DE
KU Leuven , Declarative Languages and Artificial Intelligence Group <i>Host: Prof. Bettina Berendt</i>	Sep - Nov 2022 Leuven, BE
Generali , Advanced Analytics Team <i>Host: Antonio Bencini Farina</i>	Sep - Dec 2021 Milan, IT

EXPERIENCE

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|---|-----------------------------------|
| KU Leuven, Computer Science Department, DTAI Lab
Postdoctoral Researcher (Mentors: Jesse Davis , Wouter Verbeke) | Nov 2024 - Feb 2026
Leuven, BE |
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- Led research project at the intersection of user intent modeling over time and time series change point detection methods, with applications to the KBC Group
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| University of Pisa, Computer Science Department, KDD Lab
Research Fellow | Nov 2020 - Oct 2024
Pisa, IT |
|--|---------------------------------|
- Worked on algorithmic discrimination detection methods based on fair counterfactual reasoning, including a set of deliverables for the European Commission
- | | |
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| Deloitte
Senior Consultant in Risk Modeling | May 2017 - Oct 2020
Brussels, BE |
|---|-------------------------------------|
- Designed, developed, and deployed statistical machine learning models on Python and R for Benelux clients in the energy (e.g., predictive maintenance) and financial (e.g., loss given default) sectors
- | | |
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| UK Government, Cabinet Office
Summer Economist | Jun - Aug 2016
London, UK |
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- Provided insights via statistical analysis to policy makers on a range of topics, such as studying the volunteering determinants in the UK and the variation in electoral costs across local authorities
- | | |
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| Frontera Energy
Summer Strategic & Commercial Analyst | Jun - Aug 2013
Bogota, CO |
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- Elaborated daily oil and gas price reports, providing economic insight into the effects of current events on price fluctuations in the world markets for briefings and presentations

PARTICIPATION

Invited talk, Seminario del Centro de Analítica para Políticas Públicas (CAPP)	Nov 2025
Poster, AAAI/ACM AIES	Oct 2025
Invited talk, DTAI Seminar, KU Leuven	Oct 2025
Poster, SMiLe	Jun 2025
Invited talk, Discrimination, Fairness, and Algorithms Series, University of Mainz	Apr 2025
Poster, ACM WSDM	Mar 2025
Invited talk, Causality Reading Club, University of Pisa	Jul 2024
Contributed talk, 3rd European Workshop on Algorithmic Fairness	Jul 2024
Invited talk, EAAMO Bridges, ALP Working Group	May 2024
Invited talk, ASR Group, University of Oslo	Nov 2023
Poster, ACM EAAMO	Oct 2023
Doctoral colloquium, ACM EAAMO	Oct 2023
Contributed talk, ECML Workshop on Bias and Fairness in AI	Sep 2023
Contributed talk, ACM FAccT	Jun 2023

Poster, IMS International Conference on Statistics and Data Science	Dec 2022
Poster, NeurIPS Workshop on Algorithmic Fairness through the Lens of Causality	Dec 2022
Invited talk, 1st Comète Workshop on Ethical AI, Inria Polytechnique	Sep 2022
Doctoral colloquium, ACM FAccT	Jun 2022
Contributed talk, 1st European Workshop on Algorithmic Fairness	Jun 2022

SERVICE

2026	ECML, AAAI (Reviewer) 4th ECML Workshop on Hybrid Human-ML and Decision Making (Organizer)
2025	AISTAT, ICML, ECML BIAS Workshop (Reviewer) 4th European Workshop on Algorithmic Fairness (Area Chair)
2024	NeurIPS, ACM FAccT, ICLR, ICML, ECML, ACM AIES (Reviewer) 3rd European Workshop on Algorithmic Fairness (Area Chair, Social Media Chair)
2023	NeurIPS, ACM FAccT, NeurIPS AFT Workshop (Reviewer) 2nd European Workshop on Algorithmic Fairness (Organizer, Social Media Chair)
2022	TheWebConf, NeurIPS AFCEP Workshop (Reviewer) 2nd ACM EAAMO (Social Media Chair)

European Association for Algorithmic Fairness, *Founding Board Member* [\[Link\]](#) 2023 - 2025
Supervised the yearly European Workshop on Algorithmic Fairness

The TSEconomist: Student Magazine, *Deputy Editor* [\[Link\]](#) 2016 - 2017
Led a team of 20 copy editors and writers, producing three issues

SKILLS

Programming	Python, R, PySpark; past experience in Scala
Tools	Git, L ^A T _E X, Bash; past experience in Docker, Stata, QGIS
Languages	English (fluent), Spanish (native), Italian (working proficiency), French (limited proficiency)

Last updated: August 6, 2026